Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology
Department of Electrical & Computer Engineering

Course Title
Digital Techniques Sessional

Course No: ECE 2112

Lab No: 02

Report Title
Boolean Function Simplification Using Boolean Algebra and Karnaugh Map, and Verification Using Logic Circuit Simulation

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1 Objective

The main objective of this experiment is to simplify the given Boolean functions using Boolean algebra and Karnaugh map. The simplified expressions can be used to implement logic circuits with fewer logic operations.

2 Theory

Boolean algebra provides a systematic method for simplifying logical expressions. Different laws such as the distributive law, absorption law, complement law, and De Morgan's theorem are used for this purpose.

Some useful Boolean identities are

$$A + AB = A$$

$$A(A + B) = A$$

and

$$A + A' = 1, \qquad AA' = 0.$$

De Morgan's theorems are

$$(A + B)' = A'B'$$

and

$$(AB)' = A' + B'.$$

A Karnaugh map, or K-map, is a graphical method used to simplify Boolean functions. Adjacent cells containing 1's are grouped to obtain a simpler expression.

3 Tasks

The given Boolean functions were simplified using the appropriate method. A Karnaugh map was used for the function where it provides a convenient simplification, while Boolean algebra was used for the remaining expressions.

3.1 Task 1

Given,

$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

The corresponding minterms are

$$F(A, B, C) = \Sigma m(3, 5, 6, 7)$$

A three-variable Karnaugh map is used to simplify the function.

Table 1: Three-variable Karnaugh Map for Task 1

A \ ***BC***	00	01	11	10
0	0	0	1	0
1	0	1	1	1

From the K-map grouping,

$$F = AB + AC + BC$$

Therefore,

$$\boxed{F = AB + AC + BC}$$

Output Expression

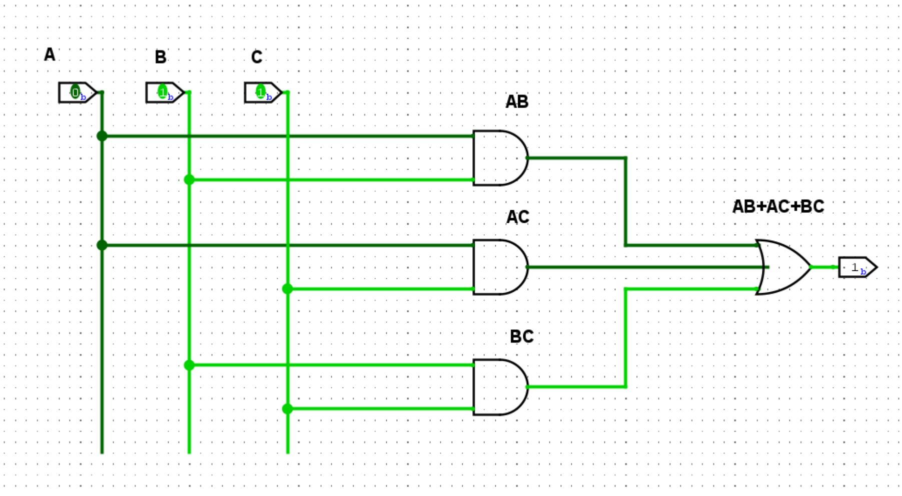

Figure 1: Output obtained for Task 1.

3.2 Task 2

Given,

$$F(A, B, C) = A(A + B)(A + B + C)$$

Using the distributive and absorption laws,

$$\begin{aligned} F &= (A + AB)(A + B + C) \\ &= A + AB + AC + AB + AB + ABC \\ &= A + AB + AC + ABC \\ &= A + AB + AC \\ &= A(1 + B + C) \\ &= A \end{aligned}$$

Therefore,

$$\boxed{F = A}$$

Output Expression

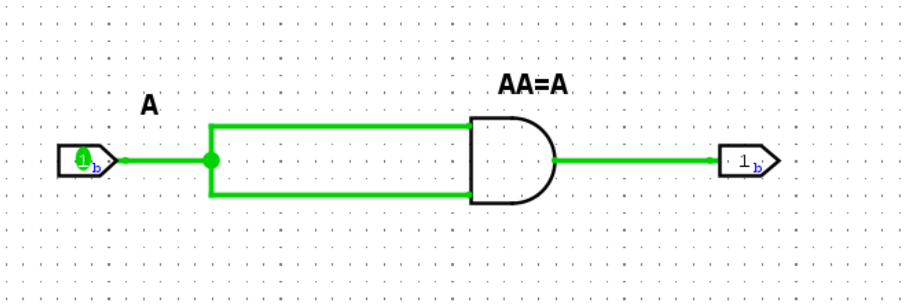

Figure 2: Output obtained for Task 2.

3.3 Task 3

Given,

$$F(A, B, C) = (A + (BC)')'(AB + ABC)$$

Since,

$$AB + ABC = AB$$

we get

$$\begin{aligned} F &= AB(A + (BC)')' \\ &= AB(A + B' + C')' \\ &= AB(A'BC) \\ &= 0 \end{aligned}$$

Therefore,

$$\boxed{F = 0}$$

Output Expression

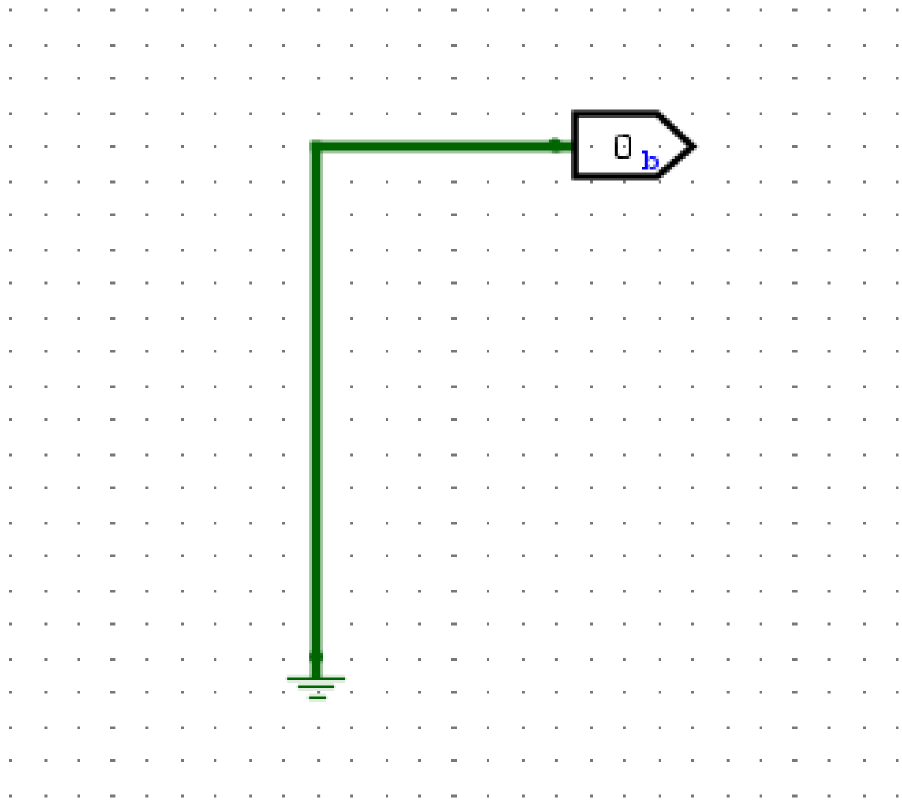

Figure 3: Output obtained for Task 3.

3.4 Task 4

Given,

$$F(A,B,C) = (B'(A+B) + (A+B)(A+B'))\,B'$$

Expanding the terms,

$$\begin{aligned} F &= (AB' + A + AB' + AB)B' \\ &= (A + AB + AB')B' \\ &= A(1 + B + B')B' \\ &= AB' \end{aligned}$$

Therefore,

$$\boxed{F = AB'}$$

Output Expression

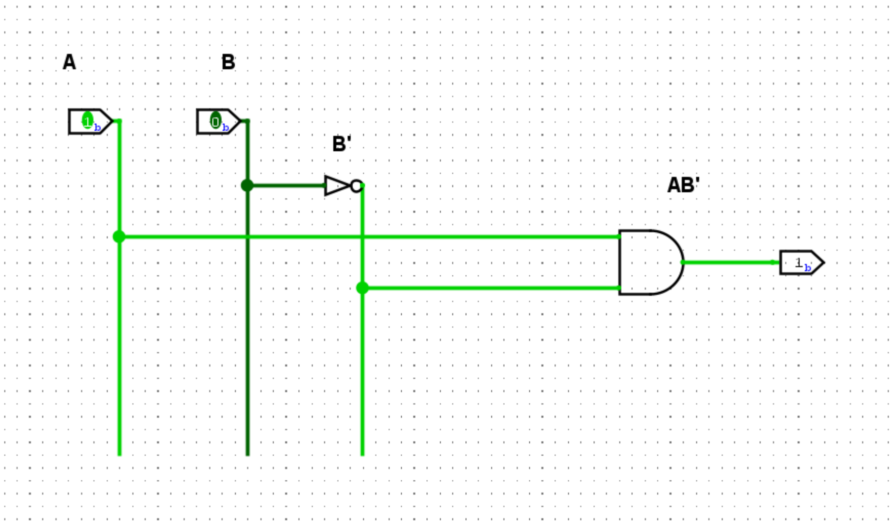

Figure 4: Output obtained for Task 4.

4 Result

The given Boolean functions were successfully simplified using Boolean algebra and Karnaugh map. The final simplified expressions obtained from the four tasks are

$$F_1 = AB + AC + BC$$
$$F_2 = A$$
$$F_3 = 0$$
$$F_4 = AB'$$

The simplified expressions are less complex than their original forms and are more convenient for logic circuit implementation.

5 Discussion

In this experiment, different Boolean algebraic laws were used to simplify the given expressions. A Karnaugh map was used for the first function because the minterms can be grouped conveniently to obtain a simpler expression.

For the remaining functions, direct Boolean algebra was sufficient. The simplification process showed how unnecessary terms can be eliminated while keeping the logical function unchanged.

6 Precautions

- Boolean variables and their complements should be written carefully.
- Each algebraic step should be checked before proceeding to the next step.
- In the K-map, only adjacent cells should be grouped.

- K-map groups should contain powers of two cells whenever possible.
- The final expression should be checked with the original expression to avoid simplification errors.

7 Conclusion

The given Boolean functions were successfully simplified using Boolean algebra and Karnaugh map. The results show that a complex Boolean expression can often be reduced to a simpler form without changing its logical operation. This type of simplification is useful for designing digital circuits with fewer logic gates.

8 References

1. M. M. Mano and M. D. Ciletti, *Digital Design*, 5th ed., Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed., Pearson, 2011.